

Epistemic Uncertainty Quantification For Pre-Trained Neural Networks

Hanjing Wang and Qiang Ji, CVPR 2024

발표자: 박민제

01 Background

Background

■ Predictive Uncertainty Quantification

- “예측 값이 틀릴 가능성은 어느 정도인가? 그 가능성은 무엇으로부터 기인하는가?”
 - Aleatoric uncertainty: 데이터의 noise로 인한, 줄일 수 없는(irreducible) uncertainty.
 - Epistemic uncertainty: 지식의 부재로 인한, 학습을 통해 줄일 수 있는(reducible) uncertainty.
- Uncertainty quantification(UQ)의 중요성
 - Trustworthiness: 예측 결과가 틀릴 가능성(confidence)을 사용자에게 알려줄 수 있다.
 - Safety: 틀릴 가능성이 높은 예측 결과(e.g., out-of-distribution)를 사전에 배제함으로써 오류로 인한 비용을 줄일 수 있다.
 - Active learning: 학습이 부족한 영역을 파악함으로써 학습을 효율적으로 수행할 수 있다.

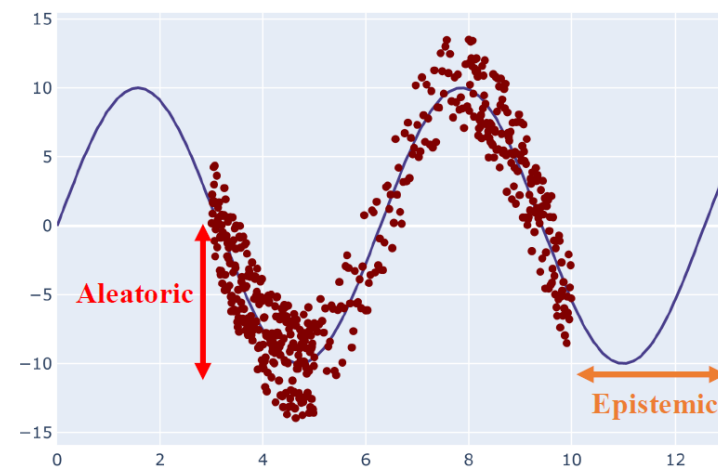


Fig. 1: A schematic view of main differences between aleatoric and epistemic uncertainties.

Background

- Bayesian Neural Networks for Epistemic Uncertainty Quantification
 - 모델이 stochastic하게 작동하도록 설계하여, 출력 값에 대한 posterior distribution를 학습하는 방법.
 - Pros: 이론적 배경(e.g., information theory, Bayesian statistics)이 탄탄한 방법이다.
 - Cons: stochasticity를 주기 위한 추가적인 조작이 필요하다. 즉, pre-trained network를 활용하기에 적절하지 않다.

Background

■ Model Sensitivity as an Epistemic Uncertainty Quantification

- 학습된 모델의 민감도 / 불안정성을 epistemic uncertainty의 surrogate로써 활용.
 - Intuition: 모델이 불안정하게 작동한다 $\rightarrow$ 학습이 부족한 영역의 데이터이다 $\rightarrow$ uncertainty가 크다.
 - Pros: 추가적인 장치 없이도 간단히 pre-trained network에 적용 가능하다.
 - Cons: 이론적으로 뒷받침되지 않은 휴리스틱 방법론이며, 다소 성능이 뒤떨어진다.
- Perturbation-based methods: 입력 값 혹은 파라미터에 noise를 가했을 때의 출력 값 변동 크기를 관찰.
 - Parameter-perturbation: $\mathbb{E}_{\Delta x \sim N(0, \sigma^2 I)} [KL(p(y|x + \Delta x, \theta^*) \parallel p(y|x, \theta^*))]$
 - Input-perturbation: $\mathbb{E}_{\Delta \theta \sim N(0, \sigma^2 I)} [KL(p(y|x, \theta^* + \Delta \theta) \parallel p(y|x, \theta^*))]$
- Gradient-based methods: 입력 값에 대한 모델의 gradient 크기를 관찰.
 - Expected gradient norm (ExGrad): $\mathbb{E}_{y \sim p(y|x, \theta^*)} [\|\nabla_{\theta^*} \log p(y|x, \theta^*)\|]$

02 Theoretical Analysis

Theoretical Analysis

■ Epistemic Uncertainty Quantification for Pre-Trained Networks

- Mutual information (MI) as an epistemic uncertainty: 출력 값($\mathbf{y}$)과 모델 파라미터(θ) 간의 MI.
 - Intuition: MI가 크다 $\rightarrow \mathbf{y}$ 가 θ 에 크게 의존한다 $\rightarrow$ 작은 θ 에도 $\mathbf{y}$ 가 크게 변한다.

$$\mathcal{I}(y; \theta | x, \mathcal{D}) = \mathbb{E}_{p(\theta | \mathcal{D})} [\text{KL}(p(y|x, \theta) || p(y|x, \mathcal{D}))] \quad (1)$$

- Approximation: $p(y|x, \mathcal{D}) \approx p(y|x, \theta^*)$

$$U_e(x) = \mathbb{E}_{p(\theta | \mathcal{D})} [\text{KL}(p(y|x, \theta) || p(y|x, \theta^*))]. \quad (2)$$

- Reparametrization: $\theta = \theta^* + \Delta\theta$

$$\mathbb{E}_{p(\Delta\theta | \mathcal{D})} [\text{KL}(p(y|x, \theta^* + \Delta\theta) || p(y|x, \theta^*))]. \quad (3)$$

Theoretical Analysis

■ Epistemic Uncertainty Quantification for Pre-Trained Networks

- Perturbation-based UQ is approximated version of UQ in BNNs.
 - 1) Pre-trained network에 사용된 학습 데이터가 충분히 클 경우, parameter-perturbation / input-perturbation 만으로도 Bayesian posterior를 근사할 수 있다.
 - 2) 따라서, perturbation-based UQ 또한 BNN과 동일한 이론적 배경을 갖는다.

Proposition 3.1. Assume the model parameters θ^* are learned given sufficient in-distribution training data $\mathcal{D}$, i.e., $|\mathcal{D}| \rightarrow \infty$. Under mild regularity conditions (i.e., the likelihood function of θ is continuous, θ^* is not on the boundary of the parameter space), perturbing θ^* by adding a small Gaussian noise can accurately approximate the posterior distribution $p(\theta|\mathcal{D})$. For example,

$$\theta^* + \sigma\epsilon \sim \mathcal{N}(\theta^*, \sigma^2 I) \quad \text{where} \quad \epsilon \sim \mathcal{N}(0, I) \quad (4)$$

and

$$\sup_{\theta} |p(\theta|\mathcal{D}) - \mathcal{N}(\theta; \theta^*, \sigma^2 I)| \rightarrow 0 \quad (5)$$

where $\sigma \rightarrow 0$ is a small positive constant.

Proposition 3.3. Denote $f(x, \theta)$ as a neural network parameterized by θ with input x . For any small perturbation $\Delta\theta \rightarrow 0$, there exists a small perturbation Δx that fulfills

$$f(x, \theta + \Delta\theta) = f(x + \Delta x, \theta). \quad (7)$$

Conversely, for any small input-space perturbation Δx , there exists a small perturbation applied on the first layer parameters that satisfies Eq. (7). Based on Eq. (7),

$$\begin{aligned} U_e(x) &= \mathbb{E}_{\Delta\theta} [\text{KL}(p(y|x, \theta^* + \Delta\theta) || p(y|x, \theta^*))] \\ &= \mathbb{E}_{\Delta x} [\text{KL}(p(y|x + \Delta x, \theta^*) || p(y|x, \theta^*))]. \end{aligned} \quad (8)$$

Theoretical Analysis

■ Epistemic Uncertainty Quantification for Pre-Trained Networks

- Gradient-based UQ is an upper bound of perturbation-based UQ.

- 1) Gradient-based UQ가 작다 → perturbation-based UQ 또한 작다.
- 2) Gradient-based UQ 또한 perturbation-based UQ의 이론적 뒷받침을 받을 수 있다.

Proposition 3.5. *The epistemic uncertainty derived by the expected gradient norm can serve as an upper bound compared to the uncertainty produced by perturbation-based methods when the perturbations are small.*

$$\begin{aligned} & \mathbb{E}_{p(\Delta\theta)} [\text{KL}(p(y|x, \theta^*) || p(y|x, \theta^* + \Delta\theta))] \\ & \leq \sum_{c=1}^C p(y=c|x, \theta^*) \left\| \frac{\partial \log p(y=c|x, \theta^*)}{\partial \theta^*} \right\| \mathbb{E}_{p(\Delta\theta)} [||\Delta\theta||] \\ & \propto \mathbb{E}_{y \sim p(y|x, \theta^*)} \left[\left\| \frac{\partial \log p(y|x, \theta^*)}{\partial \theta^*} \right\| \right] (\text{ExGrad [11]}) \end{aligned} \quad (10)$$

where $\Delta\theta \rightarrow 0$ and $\mathbb{E}_{p(\Delta\theta)} [||\Delta\theta||]$ is independent of x .

03 Proposed Method

Proposed Method

- Extension of ExGrad

- Extension 1: root-normalized expected gradient (REGrad)

- Before (ExGrad): $\mathbb{E}_{y \sim p(y|x, \theta^*)} [\|\nabla_{\theta^*} \log p(y|x, \theta^*)\|] = \sum_{c=1}^C p(y = c|x, \theta^*) \|\nabla_{\theta^*} \log p(y|x, \theta^*)\|$
- After (REGrad): $\sum_{c=1}^C \sqrt{p(y = c|x, \theta^*)} \|\nabla_{\theta^*} \log p(y|x, \theta^*)\|$
- “... to prevent an overemphasis (overconfidence) which might come from the high probability of a certain class.”

- Extension 2: layer-selective gradients

- Before: $\left\| \frac{\partial \log p(y|x, \theta^*)}{\partial \theta^*} \right\|$
- After: $\sum_{\theta_l^* \in \theta^*} a_l \left\| \frac{\partial \log p(y|x, \theta^*)}{\partial \theta^*} \right\|$
- “Gradients from deeper layers excel at discerning classification patterns and may struggle to represent out-of-distribution samples, thus being more indicative of epistemic uncertainty.”

Proposed Method

- Extension of ExGrad

- Extension 3: gradient-perturbation integration

- Before: $\left\| \frac{\partial \log p(y|x_0, \theta^*)}{\partial \theta^*} \right\|$

- After: $\left\| \frac{1}{N+1} \sum_{i=0}^N \frac{\partial \log p(y|x, \theta^*)}{\partial \theta^*} \right\|$

- “... to produce a smoother and more informative gradient representation, addressing the issue of noisy raw gradients.”

04 Experiments

Experiments

■ Purpose

- 1) BNN, perturbation-based method, gradient-based method를 다양한 시나리오에서 실험 및 성능 확인.
- 2) Non-Bayesian method도 적절한 UQ 방법론임을 실험적으로 확인.
- 3) Gradient-based method의 개선을 통해 성능이 개선됨을 확인.

■ Settings

- Out-of-distribution(OOD) detection
- Uncertainty calibration
- Active learning

Experiments

■ Results – OOD detection

- LA(BNN)에 비교하여 Non-Bayesian 방법들(e.g., Exgrad, Perturb θ)이 나쁘지 않은 성능을 보였다.
- REGrad가 ExGrad에 비해 나은 성능을 보이며, 제안한 방법들이 성능 향상에 유의미한 영향을 미쳤음을 보였다.

Table 1. OOD detection results for AUROC (%) $\uparrow$ and AUPR (%) $\uparrow$ with epistemic uncertainty. “*” represents our method (REGrad + layer-selective + perturbation). The experiments are aggregated over three independent runs.

Method	MNIST $\rightarrow$ Omniglot		MNIST $\rightarrow$ FMNIST		C10 $\rightarrow$ SVHN		C10 $\rightarrow$ LSUN		Avg
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	
NEGrad	41.99 $\pm$ 1.75	43.49 $\pm$ 0.91	40.99 $\pm$ 3.51	44.41 $\pm$ 0.91	72.80 $\pm$ 1.27	64.47 $\pm$ 1.09	66.96 $\pm$ 2.05	58.51 $\pm$ 1.64	54.20
UNGrad	92.81 $\pm$ 0.09	90.79 $\pm$ 0.33	95.41 $\pm$ 0.48	94.91 $\pm$ 0.34	38.07 $\pm$ 9.79	46.86 $\pm$ 9.31	17.08 $\pm$ 8.56	33.52 $\pm$ 9.27	63.68
GradNorm	16.52 $\pm$ 1.47	35.47 $\pm$ 0.61	22.71 $\pm$ 2.01	40.88 $\pm$ 0.61	10.73 $\pm$ 3.09	32.04 $\pm$ 1.15	10.57 $\pm$ 3.93	32.04 $\pm$ 0.89	25.12
Exgrad	97.55 $\pm$ 0.05	96.99 $\pm$ 0.16	98.11 $\pm$ 0.14	97.98 $\pm$ 0.16	88.87 $\pm$ 0.01	84.85 $\pm$ 1.15	88.23 $\pm$ 0.11	82.26 $\pm$ 1.02	91.86
Perturb x	97.16 $\pm$ 0.00	95.65 $\pm$ 0.16	97.09 $\pm$ 0.14	95.39 $\pm$ 0.31	92.35 $\pm$ 0.22	91.57 $\pm$ 1.32	82.89 $\pm$ 0.40	72.35 $\pm$ 0.55	90.56
Perturb θ	97.64 $\pm$ 0.09	97.05 $\pm$ 0.24	97.99 $\pm$ 0.16	97.64 $\pm$ 0.22	89.31 $\pm$ 1.26	85.80 $\pm$ 3.96	88.33 $\pm$ 0.11	82.99 $\pm$ 0.27	92.09
MC-AA	91.23 $\pm$ 0.35	89.40 $\pm$ 0.28	95.28 $\pm$ 0.44	95.24 $\pm$ 0.19	87.56 $\pm$ 0.57	82.22 $\pm$ 2.08	82.96 $\pm$ 0.84	71.79 $\pm$ 1.62	86.96
Inserted Dropout	97.52 $\pm$ 0.14	96.88 $\pm$ 0.35	97.00 $\pm$ 0.68	95.75 $\pm$ 0.79	88.84 $\pm$ 1.02	88.01 $\pm$ 1.19	88.18 $\pm$ 0.62	86.93 $\pm$ 0.73	92.39
Entropy	97.70 $\pm$ 0.08	97.38 $\pm$ 0.19	98.04 $\pm$ 0.22	97.94 $\pm$ 0.17	88.86 $\pm$ 0.34	84.10 $\pm$ 0.36	89.84 $\pm$ 0.30	87.72 $\pm$ 0.57	92.70
ExGrad V Term	97.70 $\pm$ 0.02	97.39 $\pm$ 0.13	98.04 $\pm$ 0.14	97.95 $\pm$ 0.26	89.04 $\pm$ 0.43	84.64 $\pm$ 0.41	89.96 $\pm$ 0.18	88.09 $\pm$ 0.29	92.85
LA	97.92 $\pm$ 0.14	97.45 $\pm$ 0.29	97.95 $\pm$ 0.27	97.37 $\pm$ 0.43	91.86 $\pm$ 0.27	88.06 $\pm$ 0.27	86.81 $\pm$ 0.27	85.04 $\pm$ 0.27	92.81
REGrad*	98.19 $\pm$ 0.02	97.95 $\pm$ 0.06	98.78 $\pm$ 0.14	98.79 $\pm$ 0.07	92.32 $\pm$ 0.59	89.42 $\pm$ 2.81	91.11 $\pm$ 0.44	89.93 $\pm$ 0.69	94.56

Method	SVHN $\rightarrow$ C10		SVHN $\rightarrow$ C100		C100 $\rightarrow$ SVHN		C100 $\rightarrow$ LSUN		Avg
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	
NEGrad	30.26 $\pm$ 1.80	39.13 $\pm$ 1.31	30.93 $\pm$ 1.24	39.38 $\pm$ 0.98	44.62 $\pm$ 2.65	42.52 $\pm$ 1.96	43.99 $\pm$ 2.21	42.74 $\pm$ 3.28	39.20
UNGrad	68.28 $\pm$ 3.45	69.21 $\pm$ 3.18	68.34 $\pm$ 2.77	68.95 $\pm$ 2.42	44.34 $\pm$ 7.36	43.44 $\pm$ 6.66	47.46 $\pm$ 4.63	45.41 $\pm$ 5.36	56.93
GradNorm	24.49 $\pm$ 3.75	36.56 $\pm$ 2.06	25.62 $\pm$ 2.72	37.26 $\pm$ 1.42	10.67 $\pm$ 2.08	32.11 $\pm$ 0.44	27.53 $\pm$ 3.91	36.97 $\pm$ 2.19	28.90
Exgrad	88.57 $\pm$ 0.61	85.55 $\pm$ 1.33	88.25 $\pm$ 0.62	85.21 $\pm$ 1.14	79.99 $\pm$ 1.37	72.68 $\pm$ 0.82	76.93 $\pm$ 1.64	71.00 $\pm$ 2.62	81.02
Perturb x	72.76 $\pm$ 0.36	60.60 $\pm$ 0.26	73.96 $\pm$ 0.42	62.27 $\pm$ 0.25	86.49 $\pm$ 1.62	86.76 $\pm$ 1.72	41.10 $\pm$ 1.42	44.35 $\pm$ 0.72	66.04
Perturb θ	89.01 $\pm$ 0.22	86.04 $\pm$ 0.41	88.75 $\pm$ 0.14	85.83 $\pm$ 0.21	50.99 $\pm$ 2.23	51.28 $\pm$ 3.08	39.42 $\pm$ 0.70	42.57 $\pm$ 1.07	66.74
MC-AA	71.53 $\pm$ 1.32	60.98 $\pm$ 1.67	72.75 $\pm$ 2.12	62.55 $\pm$ 2.72	54.74 $\pm$ 1.08	59.26 $\pm$ 2.49	31.61 $\pm$ 1.81	39.40 $\pm$ 2.22	56.60
Inserted Dropout	88.47 $\pm$ 1.23	83.97 $\pm$ 0.71	86.96 $\pm$ 1.09	82.40 $\pm$ 0.88	57.39 $\pm$ 1.46	55.49 $\pm$ 2.02	59.41 $\pm$ 1.61	58.30 $\pm$ 0.92	71.55
Entropy	89.64 $\pm$ 0.04	87.76 $\pm$ 2.19	89.09 $\pm$ 0.13	87.25 $\pm$ 0.06	53.38 $\pm$ 0.49	55.20 $\pm$ 0.18	42.28 $\pm$ 0.99	46.07 $\pm$ 0.39	68.83
ExGrad V Term	89.72 $\pm$ 0.08	87.90 $\pm$ 1.87	89.19 $\pm$ 0.07	87.40 $\pm$ 0.02	50.35 $\pm$ 0.42	52.66 $\pm$ 0.73	41.67 $\pm$ 1.51	45.91 $\pm$ 1.12	68.10
LA	80.73 $\pm$ 0.73	87.46 $\pm$ 1.00	88.77 $\pm$ 0.78	86.06 $\pm$ 1.12	80.09 $\pm$ 0.12	73.63 $\pm$ 0.39	76.50 $\pm$ 0.40	71.92 $\pm$ 0.48	80.65
REGrad*	91.11 $\pm$ 0.19	89.63 $\pm$ 0.37	90.37 $\pm$ 0.26	88.83 $\pm$ 0.37	88.06 $\pm$ 1.29	85.74 $\pm$ 1.45	79.11 $\pm$ 0.32	74.39 $\pm$ 0.67	85.91

Experiments

■ Results – Uncertainty Calibration

Table 2. Uncertainty calibration results for MNIST, C10, SVHN, and C100 datasets using rAULC $\uparrow$. “*” represents our method and the experiments are aggregated over three independent runs.

Method	MNIST	C10	SVHN	C100	Avg
	rAULC	rAULC	rAULC	rAULC	
NEGrad	0.143 $\pm$.055	0.598 $\pm$.039	-0.861 $\pm$.042	0.127 $\pm$.050	0.002
UNGrad	0.934 $\pm$.004	0.840 $\pm$.022	0.429 $\pm$.061	0.403 $\pm$.038	0.652
GradNorm	-2.810 $\pm$.022	-1.275 $\pm$.095	-1.176 $\pm$.026	-0.995 $\pm$.098	-1.564
ExGrad	0.985 $\pm$.001	0.893 $\pm$.016	0.868 $\pm$.003	0.841 $\pm$.017	0.897
Perturb x	0.985 $\pm$.001	0.890 $\pm$.014	0.830 $\pm$.001	0.772 $\pm$.032	0.869
Perturb θ	0.984 $\pm$.001	0.892 $\pm$.015	0.869 $\pm$.001	0.705 $\pm$.002	0.863
MC-AA	0.969 $\pm$.002	0.879 $\pm$.018	0.807 $\pm$.000	0.784 $\pm$.017	0.860
Inserted Dropout	0.973 $\pm$.003	0.891 $\pm$.005	0.851 $\pm$.006	0.783 $\pm$.003	0.874
Entropy	0.985 $\pm$.001	0.893 $\pm$.015	0.867 $\pm$.004	0.863 $\pm$.017	0.902
ExGrad V Term	0.985 $\pm$.001	0.893 $\pm$.015	0.865 $\pm$.004	0.854 $\pm$.020	0.899
LA	0.983 $\pm$.002	0.865 $\pm$.003	0.857 $\pm$.002	0.792 $\pm$.007	0.874
REGrad*	0.985 $\pm$.001	0.898 $\pm$.016	0.873 $\pm$.002	0.858 $\pm$.013	0.904

- REGrad가 가장 나은 성능을 보였으나, aleatoric uncertainty를 활용하는 다른 방법들의 성능에서 크게 벗어나지 않았다.

Experiments

■ Results – Active Learning

Table 3. Active learning results of averaged ACC $\uparrow$ and NLL $\downarrow$ across 10 acquisition cycles for all datasets . “*” represents our method.

Method	MNIST		C10		SVHN		C100		Avg	
	ACC	NLL	ACC	NLL	ACC	NLL	ACC	NLL	ACC	NLL
NEGrad	70.22 $\pm$ 3.02	1.833 $\pm$.04	39.13 $\pm$.72	2.074 $\pm$.02	65.12 $\pm$ 2.25	1.827 $\pm$.02	13.16 $\pm$.27	4.507 $\pm$.01	46.91	2.560
UNGrad	69.56 $\pm$ 2.94	1.867 $\pm$.03	38.90 $\pm$.74	2.071 $\pm$.01	67.27 $\pm$ 2.17	1.808 $\pm$.02	12.56 $\pm$.58	4.508 $\pm$.01	47.07	2.564
GradNorm	66.85 $\pm$ 3.18	1.884 $\pm$.04	39.02 $\pm$.81	2.069 $\pm$.01	65.43 $\pm$ 2.30	1.827 $\pm$.03	12.89 $\pm$.33	4.509 $\pm$.01	46.05	2.572
ExGrad	69.91 $\pm$ 2.21	1.847 $\pm$.03	39.20 $\pm$.87	2.069 $\pm$.01	66.76 $\pm$ 2.11	1.812 $\pm$.02	12.55 $\pm$.32	4.506 $\pm$.00	47.11	2.559
Perturb x	71.10 $\pm$ 3.43	1.844 $\pm$.05	38.57 $\pm$.72	2.072 $\pm$.01	64.00 $\pm$ 2.39	1.838 $\pm$.02	12.23 $\pm$.27	4.511 $\pm$.00	46.47	2.566
Perturb θ	71.44 $\pm$ 2.43	1.826 $\pm$.03	38.97 $\pm$ 1.03	2.070 $\pm$.02	66.31 $\pm$ 2.62	1.817 $\pm$.03	13.17 $\pm$.37	4.506 $\pm$.01	47.47	2.555
MC-AA	71.69 $\pm$.42	1.809 $\pm$.00	38.57 $\pm$.72	2.073 $\pm$.01	64.56 $\pm$ 1.17	1.831 $\pm$.02	12.14 $\pm$.40	4.515 $\pm$.01	46.74	2.557
Inserted Dropout	72.84 $\pm$ 0.62	1.836 $\pm$.00	38.34 $\pm$.11	2.075 $\pm$.01	66.99 $\pm$ 2.58	1.811 $\pm$.02	12.59 $\pm$.13	4.507 $\pm$.00	47.69	2.557
Entropy	70.40 $\pm$ 4.33	1.846 $\pm$.06	38.73 $\pm$.65	2.074 $\pm$.01	63.42 $\pm$ 2.86	1.845 $\pm$.03	11.57 $\pm$.41	4.521 $\pm$.01	46.03	2.572
ExGrad V Term	68.53 $\pm$ 4.40	1.881 $\pm$.04	38.66 $\pm$.67	2.071 $\pm$.01	62.32 $\pm$ 2.87	1.856 $\pm$.03	11.30 $\pm$.38	4.520 $\pm$.01	45.20	2.582
LA	62.65 $\pm$.58	1.935 $\pm$.02	37.56 $\pm$.29	2.083 $\pm$.00	63.44 $\pm$.75	1.846 $\pm$.00	12.59 $\pm$.12	4.505 $\pm$.00	44.06	2.592
REGrad*	75.31 $\pm$ 3.48	1.808 $\pm$.03	39.28 $\pm$.83	2.067 $\pm$.01	67.37 $\pm$ 2.13	1.807 $\pm$.02	13.37 $\pm$.25	4.505 $\pm$.00	48.83	2.547

- REGrad가 모든 벤치마크에서 가장 나은 성능을 보였다.

04 Conclusion

Conclusion

- Perturbation-based method / Gradient-based method의 이론적 뒷받침 증명.
 - Perturbation-based method는 BNN을 근사할 수 있다.
 - Gradient-based method는 perturbation-based method의 upper bound이다.
- Gradient-based method를 개선한 새로운 방법론 제시.
 - Root-normalized expected gradient (REGrad)
 - Layer-selective gradients
 - Gradient-perturbation integration
- 다양한 시나리오에서 proposed method의 유효성 확인
 - Out-of-distribution detection
 - Uncertainty calibration
 - Active learning

End of Document